



Project Presentation of Co-Op Project at Industry (Module-2)(22CS421)

On

MOODSYNC

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Supervised By

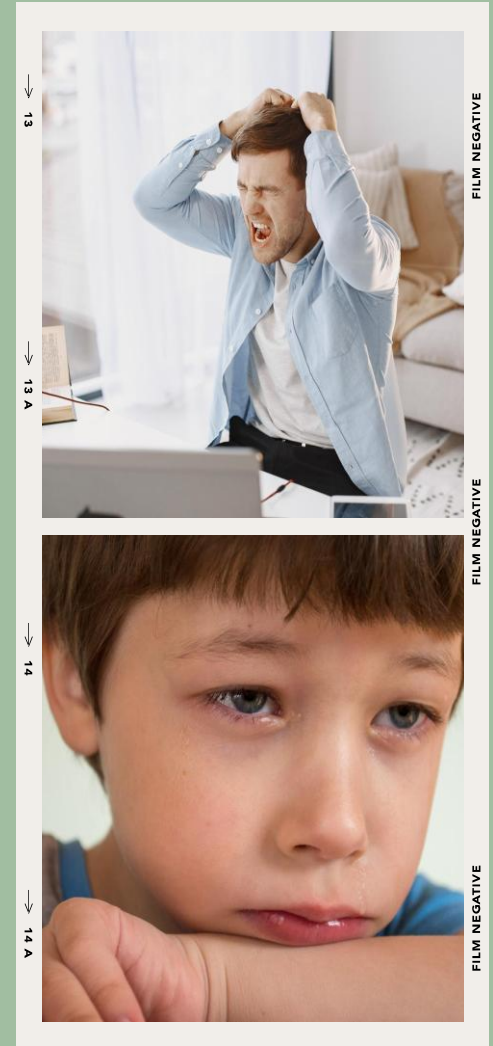
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Introduction



In today's fast-paced world, emotional well-being is often overlooked, leading to increased stress and anxiety. Studies show that 57% of people use music to manage stress. Yet, current recommendation systems rely on manual selection or predefined algorithms that don't adapt to real-time emotions. MoodSync, which is an AI-powered software bridges this gap by using facial recognition to analyze emotions and play music, enhancing emotional well-being. With applications in mental health, stress management, and emotional awareness, this innovation transforms music into personalized therapy.



Technology Stack



- Programming Language:
 - Python – The core language powering our project.
 - JavaScript (Node.js) – Powers the backend server and manages communication.
- Machine Learning & Deep Learning:
 - TensorFlow & Keras – This is for building and training deep learning models.
 - Scikit-Learn – For classical machine learning algorithms and preprocessing.
- Data Handling & Processing:
 - Pandas – For data manipulation and analysis.
 - NumPy – For numerical computing and efficient array operations.
- Computer Vision:
 - OpenCV (opencv-contrib-python) – For image processing and computer vision tasks.
- Backend & Server
 - Node.js – Server-side runtime environment to handle frontend requests and route them appropriately.
 - Express.js – Web framework for Node.js used to define routes and middleware.
 - child_process (Node.js module) – Used to run Python scripts from the Node.js backend.
- File Handling
 - Multer – Middleware for handling multipart/form-data, especially for uploading frontend images.
- Multimedia Output
 - Pygame – For audio playback during testing and model validation.
- Development & Workflow:
 - Jupyter Notebook – For interactive coding, visualization, and experimentation.
 - tqdm – For progress bar visualization to track long-running processes.
 - GitHub – For version control, collaboration, and communication.



Proposed Methodology



**LOAD THE PRE-
TRAINED AI MODEL
FOR EMOTION
DETECTION.**



**ACCESS THE WEBCAM
FOR REAL-TIME VIDEO
CAPTURE.**



**IDENTIFY FACES IN
THE VIDEO USING
OPENCV'S HAAR
CASCADE.**



**ENHANCE IMAGE FOR
BETTER AI
PROCESSING.**



**ANALYZE FACIAL
EXPRESSIONS TO
DETECT EMOTIONS.**



**SHOW DETECTED
EMOTION ON THE VIDEO
FEED.**



**CONTINUOUSLY
PROCESS FRAMES
FOR LIVE DETECTION**



CNN Model Training Workflow



**CREATE DATAFRAME
WITH EXTRACTED IMAGE
PATHS AND LABELS.**

**LOAD IMAGES,
CONVERT TO
GRAYSCALE (48X48),
NORMALIZE AND
RESHAPE FOR MODEL
INPUT.**

**DEFINE & COMPILE CNN
MODEL.**

**TRAIN THE MODEL USING
MODEL.FIT(...) ON THE
PROCESSED DATASET.**

SAVE TRAINED MODEL

**LOAD LIBRARIES & DATASET
(KERAS, PANDAS AND NUMPY)**



Emotion Detection and Music Playback Workflow



FRONTEND (HTML + JAVASCRIPT)

- THE USER SELECTS A LANGUAGE, AND A PHOTO IS CAPTURED VIA WEBCAM.
- IMAGE AND LANGUAGE ARE SENT TO [HTTP://LOCALHOST:3000/DETECT](http://localhost:3000/detect) VIA POST.

BACKEND (NODE.JS + EXPRESS)

- RECEIVES IMAGE AND LANGUAGE WITH MULTER, STORES IMAGE IN /UPLOADS.
- RUNS EMOTION.PY USING `CHILD_PROCESS.EXEC()` WITH IMAGE PATH.



EMOTION DETECTION MODEL (PYTHON + KERAS + OPENCV)

- PYTHON SCRIPT LOADS CNN MODEL AND DETECTS EMOTION FROM THE IMAGE.
- RETURNS THE EMOTION LABEL TO THE NODE.JS SERVER.

FRONTEND (AGAIN)

- FRONTEND SHOWS EMOTION AND OPENS YOUTUBE LINK IN SAME TAB.

YOUTUBE SONG RECOMMENDATION

- NODE.JS CREATES A YOUTUBE SEARCH QUERY USING EMOTION + LANGUAGE.
- SENDS THE SEARCH URL AS JSON TO THE FRONTEND.

Proposed Formulation



- Mood Sync performance depends on accurate emotion detection using CNN.
- Facial recognition accuracy is influenced by training data quality and model robustness.
- Good image quality, proper lighting, and facial alignment improve prediction results.
- User engagement increases through real-time music recommendations and user-friendly interface.
- Multilingual support enhances accessibility for diverse users.
- Effectiveness is measured through user satisfaction, listening duration, repeat usage, and low skip rate.
- Emotion detection is the core component, as wrong predictions can reduce recommendation quality.
- Optimizing detection accuracy improves reliability, user experience, and acceptance of the system.



Project Evolution Output



MoodSync

Your emotions, your music 🎧

Discover music that matches your mood in real time. MoodSync uses AI to understand your facial expressions and instantly plays music tailored to how you feel. Just start your camera, pick your language, and enjoy a personalized listening experience.

Select Language:

English

Start Camera

Detect Emotion

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Select Language:

English



Detecting...

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Results

- The system was able to identify the following emotions – **happiness, sadness, anger, calmness, surprise and neutrality.**
- The system can **accurately map emotions to music.** Eg. the detection of happy emotion led to the playback of upbeat music.
- The system takes only **45 seconds** between detection and playback of music, showcasing effective time performance.
- Further testing results showed that out of 10 samples, the system shows **accurate detection in eight of the ten samples.**
- Following are the test results.

ACTUAL EMOTION	PREDICTED HAPPY	PREDICTED SAD	PREDICTED ANGRY	PREDICTED NEUTRAL	TOTAL
HAPPY	2	0	0	0	2
SAD	0	2	0	0	2
ANGRY	0	1	1	0	2
NEUTRAL	0	0	1	1	2
TOTAL	2	3	2	1	10

Conclusion

With a powerful CNN at its core, EmoVerse model transforms raw facial expressions into real-time emotion detection. This breakthrough bridges AI and human emotions, unlocking endless possibilities—from personalized music therapy to mental wellness solutions. The future of emotion-aware technology starts here!



- Dataset Link: <https://www.kaggle.com/datasets/ananthu017/emotion-detection-fer>
- Jaiswal, A., Raju, A.K. and Deb, S., 2020, June. Facial emotion detection using deep learning. In 2020 international conference for emerging technology (INCET) (pp. 1-5). IEEE.
- Chutia, T. and Baruah, N., 2024. A review on emotion detection by using deep learning techniques. Artificial Intelligence Review, 57(8), p.203.
- Pereira, R., Mendes, C., Ribeiro, J., Ribeiro, R., Miragaia, R., Rodrigues, N., Costa, N. and Pereira, A., 2024. Systematic review of emotion detection with computer vision and deep learning. Sensors, 24(11), p.3484.
- Halkiopoulos, C., Gkintoni, E., Aroutzidis, A. and Antonopoulou, H., 2025. Advances in neuroimaging and deep learning for emotion detection: A systematic review of cognitive neuroscience and algorithmic innovations. Diagnostics, 15(4), p.456.



Thank You